

Remaining Project Plan: From SSI Validation to Floe Parameter Recovery

Summary for supervision discussion

This note lays out the finalised plan for the remainder of this project, from the current validated state through to the actual deliverable: recovering physical floe properties from simulated buoy sensor data. Green boxes summarise what is already validated and complete; blue boxes describe planned work; red boxes flag genuine risks, open questions, or scope caveats that should be read as part of the plan, not afterthoughts.

1 Current status

Forward Model (Phase 1): complete. The eigenfunction matching method (EMM) for a single floating elastic disc is implemented, validated end to end (including a machine-precision self-consistency check across the entire matching/scattering pipeline), and extended to irregular JONSWAP-forced sea states via linear superposition.

SSI algorithm and deterministic validation (Phase 2, Levels 1-2C): complete. The SSI-DATA algorithm is implemented and unit-tested function by function. A staged, increasingly demanding validation ladder against genuine Forward Model physics (Levels 1-3B) established that SSI recovers known frequencies and spatial responses to machine precision under idealised, noise-free conditions, characterised the method's frequency resolution and amplitude-sensitivity limits, and (Level 2C) used an exhaustively-verified observability-Gramian study to determine, and validate the transferability of, a defensible four-sensor placement strategy.

What remains untested. Every result to date is deterministic: no measurement noise, no sensor imperfections, only forced (not free-vibration) response, and only two-tone (not broadband) forcing. Most importantly, **the actual inverse mapping from identified dynamics back to physical floe parameters has not yet been built or tested at all.** This plan is organised specifically to close these gaps while protecting the time needed for that final, central piece.

2 Conceptual structure: four separate questions, not one ladder

A central clarification underlies this plan: several previously-conflated questions are genuinely independent, and should be validated as such.

Question	Answered by
Is the Forward Model itself physically correct?	An <i>independent</i> , differently-formulated model of the same floe (Section 4) - not yet attempted; every check so far is self-consistency, not independent cross-validation.
Does SSI correctly recover known dynamics from data?	The Level 1-8 validation ladder (Sections 3-5), using synthetic data with known ground truth.
Can SSI's output be mapped back to physical floe parameters?	The inverse-mapping twin test (Section 7) - the actual project deliverable.
Does the whole pipeline survive realistic conditions?	Levels 5-7 (noise, sensor imperfections, broadband forcing), layered on top of the above once each is independently validated.

3 Stage 1: robustness, Levels 4-6

Level 4 (phase variation). All validation to date has used zero or arbitrary fixed phases between components. Confirming recovery remains robust once phases are genuinely varied is cheap, using only already-existing machinery.

Level 5 (additive noise, multiple SNRs and seeds). A real IMU noise characterisation is being carried out this week, directly informing the noise levels and statistical structure used here, rather than an assumed or arbitrary noise model. This is the first genuinely statistical (not purely deterministic) validation stage in the project, and is expected to surface real robustness limits, e.g. around the still-unexplained missing-mode result from earlier hand-built noisy tests, and around the known context-dependence of the `classifyPoleStability.m` frequency-matching tolerance.

Level 6 (sensor imperfections). Gain error, bias, and a failed/degraded channel, tested using the same multi-sensor infrastructure already built for Level 2C.

4 Stage 2: independent Forward Model validation via Capytaine

Why this is a genuinely new validation category. Every check performed on the Forward Model to date is *self-consistency* validation: confirming the EMM’s own equations are solved correctly (closed-form identities, convergence studies, the free-edge residual check). None of them confirm the EMM’s underlying physics is correct by comparison against an independently-formulated model. This stage closes that gap.

4.1 Method

A structural modal basis for the plate is combined with hydrodynamic coefficients from Capytaine, a frequency-domain boundary-element (BEM) solver, to build an independent wet frequency-response function (FRF) for direct comparison against the EMM’s own FRF (the same FRF-sweep procedure used for the inverse mapping, Section 7).

Structural basis: analytical, not a full finite-element model. Given the floe’s simple circular geometry, classical free-circular-plate mode shapes and natural frequencies (Leissa, *Vibration of Plates*) are used directly, avoiding the setup risk and time cost of standing up finite-element software for comparatively little physical benefit over the analytical result.

Hydrodynamics: Capytaine, with generalised (elastic) degrees of freedom. Capytaine computes frequency-dependent added mass $A(\omega)$ and radiation damping $B(\omega)$ directly. Crucially, it supports user-defined generalised body modes, allowing the structural plate modes themselves to be used as the hydrodynamic degrees of freedom, rather than restricting the comparison to rigid-body heave/pitch/roll only - the latter would not exercise the flexural physics this project’s parameter recovery actually depends on.

Coupled model. For structural modal coordinates q , the coupled frequency-domain system is

$$\left[K_s + K_h - \omega^2 (M_s + A(\omega)) + i\omega B(\omega) \right] q = F_{\text{exc}}(\omega), \quad (1)$$

where M_s, K_s come from the structural (Leissa) model and $A(\omega), B(\omega)$ from Capytaine. Sweeping ω gives an independent wet FRF, directly comparable to the EMM’s own.

Known caveat: hydrostatic stiffness for generalised DOFs. Capytaine’s documentation notes approximations in its automatic hydrostatic stiffness calculation specifically for generalised (non-rigid-body) degrees of freedom. This will be handled deliberately - computed independently rather than trusted from an automatic routine - rather than risking a silent error in K_h contaminating the whole comparison.

4.2 Validation criterion

Resonance structure (peak locations and approximate widths) of the two independently-derived FRFs are compared directly. Agreement supports the Forward Model’s underlying physical correctness from a source entirely outside its own equations; disagreement would itself be a valuable, reportable finding, pointing to which physical effect (added mass, radiation damping, or something in the EMM’s own formulation) is responsible.

This stage is explicitly timeboxed. The generalised-DOF Cappytaine setup is the one genuinely unfamiliar, uncertain step in this entire plan. If it proves substantially harder than anticipated, this stage is scoped down or dropped before it is allowed to consume time needed for Section 7, the actual deliverable.

5 Stage 3: Level 8, true modal validation

5.1 Why Level 8 is placed before Level 7

Level 8 asks a clean question: given data generated from *known* natural frequencies and damping ratios (a genuine free-vibration signal, not a forced one), does SSI recover them? Level 7 (broadband JONSWAP forcing, Section 6) is harder to interpret in isolation, because JONSWAP forcing is coloured, not white: a peak in the resulting response can reflect the forcing spectrum’s own shape as much as the floe’s genuine dynamics (the same distinction underlying this project’s original choice of SSI over ERA-NExT for exactly this reason). Validating clean modal recovery first, before this confound is introduced, gives a meaningful baseline to interpret Level 7’s harder result against.

5.2 Two possible sources of ground truth, in order of preference

Preferred: the coupled wet model from Stage 2. If the Cappytaine-based coupled model (Section 4) is working, it provides $M_s, K_s, A(\omega), B(\omega)$ for a genuinely independent wet dynamical model - independent specifically in the sense that SSI would be tested against data it played no part in generating, unlike testing it against numbers extracted from the very model (the EMM) being validated.

The damping caveat, and the pragmatic compromise. With frequency-dependent radiation damping $B(\omega)$, there is no single constant damping ratio in the way a simple Rayleigh-damped structural model would have. The fully rigorous time-domain equation of motion is a Cummins-type convolution,

$$(M_s + A_\infty)\ddot{q} + Kq + \int_0^t K_r(t - \tau)\dot{q}(\tau) d\tau = 0, \quad (2)$$

which Cappytaine’s frequency-domain output does not directly solve. Given the project timeline, a full radiation-memory time-domain solver will **not** be attempted. Instead, a reduced model with hydrodynamic coefficients frozen near each resonance is used to approximate a genuine free-decay signal, explicitly documented as an approximation, not the fully rigorous Cummins solution. Upgrading to the full convolution form remains a stated, lower-priority stretch goal only if time allows.

Fallback: an approximate decay synthesised from the EMM’s own FRF. If Stage 2 does not complete in time, a simple damped-sinusoid decay is built directly from $(\omega_{\text{res}}, \zeta_{\text{HP}})$ already extracted from the EMM’s own FRF sweep (Section 7). This is a weaker test, since the ground truth is not independent of the model being validated, but it is fast, low-risk, and was the original fallback identified before Stage 2 was scoped.

5.3 Validation criterion

Synthetic sensor histories are generated from whichever wet free-decay model is available, at sensor locations informed by (though not blindly copied from) the Level 2C placement result. SSI is run on this data with no knowledge of how it was generated, and its recovered $(\hat{\omega}_n, \hat{\zeta}, \hat{\Phi})$ are compared directly against the known generating values, including spatial mode-shape MAC where multiple sensors allow it.

6 Stage 4: Level 7, realistic broadband forcing

Once Level 8 has established a clean modal-recovery baseline, the same machinery is applied under genuine JONSWAP forcing (already fully implemented in this project’s Forward Model extension), asking whether SSI can still recover the floe’s structural dynamics despite coloured, statistically realistic forcing - the condition closest to what a real deployed buoy would actually experience.

7 Stage 5: the inverse mapping (the project deliverable)

7.1 The forward prediction: an FRF sweep, not a direct eigenvalue solve

For a trial parameter set (β, γ, R) , the already-existing, unmodified Forward Model is swept finely over the non-dimensional frequency α to build a frequency-response function $H(\omega)$ (response amplitude and phase at chosen sensor locations, as a function of forcing frequency) - exactly analogous to a classical experimental modal test, but computed rather than measured.

Precise statement of what this actually estimates. The mapping is

$$(\beta, \gamma, R) \longrightarrow H(\omega) \longrightarrow (\omega_{\text{res}}, \zeta_{\text{HP}}, \Phi_{\text{res}}), \quad (3)$$

where ω_{res} is a resonance peak location and ζ_{HP} a half-power bandwidth estimate. For a lightly-damped, well-isolated single-degree-of-freedom peak, these closely approximate the true modal natural frequency and damping ratio - but in a wet, multi-degree-of-freedom system with frequency-dependent hydrodynamic coefficients, they are not automatically identical to the true modal (ω_n, ζ) that SSI itself extracts as genuine poles of a fitted state-space model. This distinction is tracked explicitly throughout this stage rather than assumed away; Stage 2's independent FRF, if available, can help quantify how good this approximation actually is for this floe.

7.2 Identifiability check: required before any optimisation is trusted

Why this step cannot be skipped. An optimiser will converge to *some* answer and report a small residual regardless of whether the underlying parameters are actually distinguishable from the available modal signature. If two of β, γ, R affect the predicted (ω_n, ζ, Φ) in nearly the same way, an optimiser can report an excellent-looking fit to the *wrong* physical parameters - a failure mode considerably more dangerous than a slow or poorly-converging optimiser, since it would look like success until independently checked. This directly follows up on an identifiability concern flagged early in this project (recovering flexural rigidity from heave-dominated data alone was already suspected to be structurally weak).

Before any optimisation is attempted, the local sensitivity (Jacobian) matrix

$$J = \frac{\partial(\omega_1, \omega_2, \zeta_1, \zeta_2, \Phi, \dots)}{\partial(\beta, \gamma, R)} \quad (4)$$

is evaluated numerically (via finite differences on the existing forward-model sweep, no new machinery required) at a representative trial point. Its conditioning is inspected directly: near-singularity, or two columns pointing in nearly the same direction, indicates a genuine identifiability limitation for that parameter combination, to be reported honestly rather than papered over by an optimiser that happens to converge anyway.

7.3 Optimisation and the twin test

Subject to the identifiability check above, the inverse problem is posed as

$$\min_{\beta, \gamma, R} \|f(\beta, \gamma, R) - \hat{f}_{\text{SSI}}\|^2, \quad (5)$$

where $f(\beta, \gamma, R)$ is the FRF-sweep prediction above and $\hat{f}_{\text{SSI}}$ is SSI's actual identified modal signature from synthetic sensor data. The success criterion, mirroring this project's twin-test discipline throughout, is a genuine recovery test:

$$(\beta^*, \gamma^*, R^*) \longrightarrow \text{synthetic sensor data} \longrightarrow \text{SSI} \longrightarrow \text{inverse fit} \longrightarrow (\hat{\beta}, \hat{\gamma}, \hat{R}) \stackrel{?}{\approx} (\beta^*, \gamma^*, R^*). \quad (6)$$

Level 2C sensor placement: a validated starting point, not a universal answer. The Level 2C λ_{min} -optimal arrangement was derived for one specific, normalised frequency pair (2.70/6.00 rad/s). If a trial (β, γ, R) substantially changes the underlying spatial response fields, the optimal

geometry can change with it. This configuration is used as the sensor layout for the *first* twin test, as a reasonable and already-validated starting point - not claimed to be optimal for every future parameter combination this stage may explore.

8 Scheduling and risk management

Start the inverse-mapping scaffolding now, in parallel with Stage 1. The identifiability (sensitivity Jacobian) analysis needs nothing beyond the already-existing, unmodified Forward Model swept over a small grid of trial parameters - it has no dependency on noise characterisation, Capytaine, or any other stage in this plan, and can begin immediately.

Priority ordering if time becomes constrained. The inverse-mapping twin test (Section 7) is the actual project deliverable and takes priority over every other stage. Stage 2 (Capytaine) is explicitly timeboxed and will be scoped down or dropped first if needed. Stage 3 (Level 8) falls back to the FRF-derived approximate decay if Stage 2 does not complete. Levels 4-7 are valuable robustness demonstrations but are not, on their own, the project's central claim.

9 Summary

Stage	What it establishes	Priority if time-constrained
1: Levels 4-6	Robustness under phase variation, real-noise SNR/seeds, sensor imperfections	Keep; cheap and already-tooled
2: Capytaine FM validation	Independent cross-validation of the Forward Model itself	Timeboxed; drop first if needed
3: Level 8	Clean modal-recovery validation (SSI vs. known wet poles)	Fall back to FRF-derived decay if Stage 2 unavailable
4: Level 7	Recovery under realistic broadband (JONSWAP) forcing	Keep; existing machinery only
5: Inverse mapping	The project deliverable: parameter recovery, gated by an identifiability check	Highest priority throughout; scaffolding starts immediately